Venkata Lakshmi Parimala Pasupuleti

Sr Software Engineer · Distributed Systems · Real-Time Systems · Experimentation Platform · A/B Testing · Python · Go · Java · Spark · Kafka · CI/CD · Production Engineering

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PROFESSIONAL SUMMARY

Software engineer with 9+ years of experience building distributed systems, large-scale data pipelines, and production backend services in Python, Java, and Go. Currently at Intuit building A/B experimentation frameworks for LLM provider comparison (AgentProbe), leveraging PySpark for large-scale batch pipeline optimization, and implementing feature flag-based progressive rollout for LLM features — with real-time dashboards and anomaly detection for production services. At Entrata (7+ years), owned high-throughput async batch workflows, large-scale data import pipelines (1.6M+ records), and end-to-end production features on a 100K+ MAU multi-tenant SaaS platform. Built AgentProbe, an open-source experimentation framework scoring LLM outputs across five providers using statistical evaluation methodology. M.S. Data Analytics Engineering from George Mason University (GPA 3.87, May 2026) — providing causal inference, statistical analysis, and experimental design foundations directly applicable to Uber's Experimentation Platform team.

CORE COMPETENCIES

Distributed Systems | A/B Testing & Experimentation | Causal Inference | Statistical Analysis | Experimental Design | Spark / PySpark | Batch Workflows | Real-Time Dashboards | Anomaly Detection | Feature Flags & Progressive Delivery | Python · Java · Go | Kafka · Real-Time Streaming | Presto · Pandas · Jupyter | Production Engineering | CI/CD · Docker · Kubernetes | Observability · On-Call

EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — Financial AI Platform

- **Productionize LLM services** at scale: integrate Anthropic Claude into Python/FastAPI services for real-time financial data processing, natural-language queries, and summaries — full end-to-end ownership from design through incident response.
- Leverage **PySpark for large-scale data processing and batch pipeline optimization**: feature engineering, data ingestion, and batch prediction workflows across millions of financial records.
- Implement **feature flag-based progressive delivery for LLM features** — controlled canary rollouts with instant rollback, enabling safe productionization without full redeploys.
- Run **A/B experimentation framework for LLM provider comparison** (AgentProbe): controlled experiments across Anthropic, OpenAI, and Azure OpenAI with statistical significance checks on accuracy, latency, and hallucination rate; productionize causal-inference evaluations into APIs.
- Build **real-time dashboards and anomaly detection** for production LLM service health — Prometheus/Grafana observability, OpenTelemetry tracing, automated alerting on quality degradation.
- Build ML pipelines (scikit-learn, XGBoost) from data ingestion and feature engineering through **production model deployment and prediction APIs**; containerize and ship via GitHub Actions CI/CD to AWS.
- Design prompts and structured-output schemas; add guardrails including input validation, sensitive-data handling, and fallbacks when model output fails quality thresholds.

Entrata — Apartment & Affordable Housing SaaS

Oct 2020 – Jul 2024

Senior Software Engineer — Multi-Tenant Platform (100K+ Monthly Active Users)

- Built **RabbitMQ async batch workflow** for rent updates across thousands of tenant records: high-throughput message dispatch, retry logic with exponential backoff, dead-letter queues, and idempotency guards — fault-tolerant processing at millions of events.
- Designed and owned **high-throughput data pipelines** processing millions of records per cycle (HUD income-limit imports, 1.6M+ records with validation and rollback); Python/Java/SQL stack, end-to-end ownership including incident response.
- Delivered 20+ features **end-to-end** (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints — design, build, test, production deploy, and on-call support.
- Wrote Python/Bash automation that cut manual operational work by 80%; built internal **real-time debugging and observability tools** that reduced triage time by 40%.
- Optimized 200+ SQL queries using execution plans and indexing, improving query performance by 30% on high-load production tables.
- Led **Docker microservices redesign** (50% better scalability, 30% faster deployments) and AWS S3 document migration including Libsodium encryption for resident PHI.
- Reviewed 300+ pull requests; provided technical leadership on design reviews for high-load distributed components.

Entrata India

Sep 2018 – Oct 2020

Junior Software Engineer — Property Management Platform

- Built property management modules (Python, PHP, Java, PostgreSQL) for affordable housing and rental management at scale.
- Delivered large-scale **data import and reconciliation workflows (1.6M HUD/Tax Credit records)** with validation, deduplication, and rollback safeguards.
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

PROJECTS

AgentProbe — Open-Source LLM Experimentation & Evaluation Toolkit

Open Source

Python · FastAPI · LangGraph · ChromaDB · Pandas · GitHub Actions · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama

Built a **controlled A/B experimentation framework** for LLM provider comparison: 15-endpoint FastAPI backend routes prompts across five providers; scores outputs on factual accuracy, hallucination rate, retrieval quality, and completeness using LLM-as-judge. **Statistical evaluation methodology** with golden datasets and CI regression tests — productionized causal-inference scoring into batch API workflows. Evaluated 1,240+ test cases with statistically significant accuracy comparisons across providers. github.com/venkatalakshmiparimala/AgentProbe

Leasel — Multi-Agent AI Platform with RAG

Python · LangGraph (8-agent) · FastAPI · Hybrid Retrieval (TF-IDF + Semantic) · OCR · SQLite · CI/CD

Built a lease-compliance app where an 8-agent LangGraph workflow (triage, retrieval, reasoning, validation) processes queries with **93.6% precision and 87.9% F1 across 1,240 test cases**. RAG pipeline: OCR ingestion, section-aware chunking, hybrid search with semantic reranking, answers cited to exact source. Human escalation guardrails for financial and eligibility decisions.
github.com/venkatalakshmiparimala/Lease_complaince_rag

CrashClarity — Large-Scale Statistical Analysis & ML Pipeline (104K Records)

Python · Pandas · scikit-learn · R · Jupyter · Statistical Modeling · GitHub Pages

Built a **causal analysis and ML pipeline on 104K+ UK accident records** — feature engineering, Random Forest and Logistic Regression classifiers (85%+ accuracy), **statistical significance testing, experimental design, and causal inference**. M.S. Data Analytics Engineering capstone: end-to-end from Pandas/Jupyter data prep through model evaluation to interactive dashboard — demonstrating the statistics and experimental analysis foundations directly applicable to Uber's Experimentation Platform.

EDUCATION

George Mason University — **M.S. Data Analytics Engineering** May 2026

GPA 3.87 · Statistics, Causal Inference, Experimental Design, Machine Learning, Data Engineering · Fairfax, Virginia

JNTU Anantapur — **B.Tech Computer Science & Engineering** 2017

TECHNICAL SKILLS

Languages: Python (9+ yrs), Java (9+ yrs), Go, Scala, TypeScript, JavaScript, PHP, SQL, Bash

Data & Analytics: PySpark, Spark, Presto, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, Jupyter, R

Experimentation & Statistics: A/B Testing, Causal Inference, Statistical Analysis, Experimental Design, Feature Flags, Progressive Delivery, Anomaly Detection, Batch Workflows

Distributed & Real-Time Systems: Kafka, RabbitMQ, Celery, caching, sharding, fault tolerance, high-throughput pipelines, event-driven architecture

Backend & APIs: FastAPI, Flask, Django, Spring Boot, Node.js, REST APIs, GraphQL, microservices

AI & ML Production: LLM APIs (Anthropic Claude, OpenAI, Azure OpenAI, Groq), ML model deployment, LangGraph, LangChain, RAG pipelines, LLM-as-judge evaluation, MCP

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Databases: PostgreSQL, MySQL, Oracle, Redis, MongoDB, Snowflake, ChromaDB, Pinecone, Elasticsearch, ETL Pipelines

Observability: Prometheus, Grafana, OpenTelemetry, real-time dashboards, incident response, on-call, SLA ownership